AASARA (ఆసరా)

A Telugu-First AI Voice Guide to Government Welfare Schemes

Problem Statement

Many people, especially those who are more comfortable speaking their regional language, may find it difficult to discover government welfare schemes, understand eligibility requirements, and identify the documents needed to apply. Language barriers and complex information can make accessing scheme details challenging.

Proposed Solution

Aasara is an AI-powered voice assistant designed to make government scheme information easier to access through natural conversations. Users can ask questions in Telugu or Hindi and receive relevant scheme information through voice-based interaction.

Key Features

- **Voice Interaction:** Users can ask questions naturally instead of typing.
- **Speech Recognition:** Gnani AI Prisma converts spoken questions into text.
- **AI-Based Understanding:** Google Gemini helps interpret user queries and generate responses.
- **Scheme Discovery:** A rule-based eligibility engine checks user information against a dataset of 9 central and Telangana government schemes.
- **Voice Responses:** Gnani AI Timbre converts responses into spoken Telugu.
- **Follow-up Questions:** The assistant can ask for additional information to assess potential eligibility.
- **Scheme Information:** Provides available details about eligibility, required documents, application procedures, and source links.

Technology Stack

Python | Streamlit | Gnani AI Prisma v2.5 (STT) | Gnani AI Timbre v2.5 (TTS) | Google Gemini | JSON

How It Works

1. The user asks a question using their voice.
2. Gnani Prisma converts the speech into text.
3. Google Gemini helps understand the user's intent.

4. The eligibility engine checks the available scheme information.
5. The assistant generates a response and Gnani Timbre speaks it in Telugu.

Expected Impact

Aasara aims to reduce language barriers and make government welfare scheme information easier to understand and access. It provides a simple conversational interface, particularly for users who prefer voice interaction over navigating complex websites.

Current Limitations

The current prototype covers 9 schemes. Some Telangana scheme details are based on news reports and require local verification. Eligibility results are informational estimates, not official government decisions. Speech recognition accuracy has not yet been formally evaluated.

Gnani AI Integration

This project uses Gnani AI's Prisma Speech-to-Text and Timbre Text-to-Speech models to enable voice-based interaction.

Project: Aasara (ఆసరా)
Demo Video: https://lnkd.in/p/di8q_QQ7
GitHub:https://github.com/srivallika26/Aasara.git

Disclaimer: Aasara is an independent informational prototype and is not an official government service.